

André M. Silva

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Short Bio

I am André Silva, a postdoctoral researcher at the Instituto de Astrofísica e Ciências do Espaço. My research spans instrumentation and exoplanetary science. I am currently working on the PoET solar telescope, leading the development of its observation software. In parallel, I focus on the development of novel algorithms for precise radial velocity extraction and analyze systematic effects that impact such measurements. Over the past three years, I have lectured an advanced course on Python, a Master's-level course on Bayesian statistics, and a PhD-level module on high-resolution spectroscopy.

Education

PhD	Faculdade de Ciências , Doctor's Degree in Astronomy Thesis - A new paradigm for the estimation of precise stellar radial velocities	Universidade do Porto 2019 – 2024
MSc	Faculdade de Ciências , Mestrado Integrado em Engenharia Física Thesis - An expansion to the CHEOPS mission official pipeline.	Universidade do Porto 2014 – 2019

Awards

IAU PhD Prize 2024 🔗 , Division B	2025
Awarded the PhD Prize 2024 from the International Astronomical Union (IAU) Division B (Facilities, Technologies, and Data Science) for the work developed during my PhD thesis.	
EAS MERAC 2026 🔗 , Division B	2025
Awarded the European Astronomical Society (EAS) MERAC Prize in Observational Astrophysics for the work developed during my PhD thesis.	

Experience

Instituto de Astrofísica e Ciências do Espaço , Post-doctoral researcher Development of the observational software for the PoET solar telescope.	CAUP Nov 2024 – present
Faculdade Ciências da Universidade do Porto , Invited Assistant Professor Invited assistant Professor to teach a PhD-level unit on high-resolution spectroscopy	Physics department Nov 2024 – present
Faculdade de Ciências da Universidade de Lisboa , Post-doctoral researcher Development of software to measure day-time seeing, under the FIERCE ERC project	FCiências.ID Aug 2024 – Nov 2024
Faculdade Ciências da Universidade do Porto , Invited assistant Invited assistant for the practical classes of MSc-level unit on Bayesian data analysis.	Physics department Feb 2024 – Sept 2024
Instituto de Astrofísica e Ciências do Espaço , PhD fellow FCT funding for the PhD thesis "A new paradigm for the estimation of precise stellar radial velocities"	Physics department 2019 – July 2024

Talks

ENAA XXXVI (invited talk) 🔗 , PoET - A portuguese solar telescope in the Atacama desert	Sept 2026
IAU symposium 408 , Looking at the Sun through ESPRESSO - the PoET solar telescope	Aug 2026
COSPAR2026 (invited talk) , POET: LOOKING AT THE SUN TO FIND OTHER EARTHS	Aug 2026
EAS2026 (invited talk) , On the extraction of precise radial velocities from high-resolution spectra - avenues and challenges	July 2026

exoplanets6 , Intrinsic limitations to template-based radial velocity extraction from high-resolution spectra	June 2026
EPRV x Solar Physics workshop 🔗 , Looking at the Sun through ESPRESSO - The PoET solar telescope	June 2026
Physis Week 2025 , Python for data analysis	Sept 2025
ENAA XXXV 🔗 , A Quest to Detect a Star Velocity: Radial Velocity Extraction and Associated Challenges	Sept 2025
EPRV6 🔗 , A systematic bias in template-based RV extraction algorithms	July 2025
Planets throughout the Habitable Zone 🔗 , A systematic bias in template-based RV extraction algorithms	June 2025
PoET workshop 2 🔗 , 2 talks: PoET observation software; Observing strategies and their role in the improvement of RV analysis	Nov 2024
TOI III 🔗 , Approaches for RV extraction – s-BART and the first steps towards a fully Bayesian model	July 2023
EPRV 5 🔗 , Towards a fully Bayesian RV extraction model	Mar 2023
Exoplanets 4 EPRV splinter , sBART application to the ESPRESSO WG1 targets	Feb 2023
ESPRESSO science team meeting , s BART a semi Bayesian implementation of template matching for precise Radial Velocities	May 2022

Seminars

Astronomical Institute of the Czech Academy of Sciences , Avenues and challenges for radial velocity extraction – s-BART & PoET	Aug 2025
Astronomy department, Uni. of Geneva , Avenues for radial velocity extraction – s-BART & PoET	Jan 2025
ESA research seminar, European Space Agency (Madrid) , Towards an improvement in the characterisation of stellar radial velocities	May 2024
Café com Física 🔗 Departamento de Física da Universidade de Coimbra , The (radial) velocity of stars - detection and characterisation of exoplanets	Mar 2024

Supervision

A new window into stellar noise in exoplanet research - the PoET view , Co-supervisor	MsC thesis July 2026 – present
Improved modelling of CCF profiles for activity-insensitive radial-velocity extraction , Supervisor	Undergraduate project Feb 2026 – July 2026
Measuring Zeeman Broadening in Stellar Spectra , Co-supervisor	Undergraduate project Feb 2026 – July 2026
Transforming H-alpha images of the Sun in astronomical seeing for the PoET solar telescope , Supervisor	Undergraduate project Feb 2025 – July 2025
Looking at the Sun, finding other Earths - Identification of Solar Regions , Co-supervisor	MsC thesis Oct 22 - Nov 23
Development of a GUI interface for the exploitation of data from the s-BART pipeline , Co-supervisor	Undergraduate project Feb 2024 – July 2024
Solar-to-sky coordinate conversion for PoET operations , Co-supervisor	Undergraduate project Feb 2024 – July 2024

Exoplanet detection through a new model for the correction of instrumental effects, Co-supervisor	Undergraduate project Feb 2024 – July 2024
Development of a tool for the normalization of stellar spectra - application to ESPRESSO data, Co-supervisor of BII project with ref CIAAUP-03/2023-BII	Funded project Feb 2024 – July 2024
A new activity proxy for finding other Earths, Co-supervisor	Bachelors project Nov 2022 – Feb 2023
An analysis of the performance of CHEOPS mission pipelines - the DRP and archi, Co-supervisor	Bachelors project Feb 2020 – June 2020

First Author Publications

ASTRA: A Python Package for Cross-Instrument Stellar and Telluric Template Construction	2026
André M. Silva et al; (JOSS, 10.21105/joss.09413)	
A systematic bias in template-based radial velocity extraction algorithms	2025
André M. Silva et al; (A&A, 10.1051/0004-6361/202554955)	
A novel framework for semi-Bayesian radial velocities through template matching	2022
André M. Silva et al; (A&A, 10.1051/0004-6361/202142262)	
archi: pipeline for light curve extraction of CHEOPS background stars	2020
André M. Silva et al; (Monthly Notices of the Royal Astronomical Society, 10.1093/mnras/staa1443)	

Selected Co-authored Publications

Gas-depleted planet formation occurred in the four-planet system around the red dwarf LHS 1903	2026
Wilson et al; (Science, 10.1126/science.adl2348)	
A comprehensive study on radial velocity signals using ESPRESSO: Pushing precision to the 10 cm/s level	2025
Figueira et al; (A&A, 10.1051/0004-6361/202553869)	
PoET: the Paranal solar ESPRESSO Telescope	2025
Santos et al; (Published in The Messenger vol. 194, 10.18727/0722-6691/5381)	
TOI-512: Super-Earth transiting a K-type star discovered by TESS and ESPRESSO	2025
Rodrigues et al; (A&A, 10.1051/0004-6361/202452887)	
KOBE-1: The first planetary system from the KOBE survey: Two planets likely residing in the sub-Neptune mass regime around a late K-dwarf	2025
Balsalobre-Ruza et al; (A&A, 10.1051/0004-6361/202452631)	
TESS and ESPRESSO discover a super-Earth and a mini-Neptune orbiting the K-dwarf TOI-238	2024
Suárez Mascareño et al; (A&A, 10.1051/0004-6361/202348958)	
A sub-Earth-mass planet orbiting Barnard's star	2024
González Hernández et al; (A&A, 10.1051/0004-6361/202451311)	
Two temperate Earth-mass planets orbiting the nearby star GJ1002	2023
Mascareño et al; (A&A, 10.1051/0004-6361/202244991)	
An unusually low-density super-Earth transiting the bright early-type M-dwarf GJ 1018 (TOI-244)	2023
Castro-González et al; (A&A, 10.1051/0004-6361/202346550)	
A candidate short-period sub-Earth orbiting Proxima Centauri	2022

Faria et al; (A&A, [10.1051/0004-6361/202142337](https://doi.org/10.1051/0004-6361/202142337) [↗](#))

The KOBE experiment: K-dwarfs Orbited By habitable Exoplanets: Project goals, target selection, and stellar characterization

2022

Lillo-Box et al; (A&A, [10.1051/0004-6361/202243898](https://doi.org/10.1051/0004-6361/202243898) [↗](#))

Organization

Coordinator of internal working group focused on modelling time-series for exoplanet detection and characterisation

2025 - now

Organizer of the scientific seminars of CAUP

September 2025 - now

Co-organizer of PoET's Working Group 1 - Radial velocities

2025 - now

Coordinator and lecturer of advanced course "Python for astronomers"

2021 - now

SoC of Dias da Física 2025

2025

Part of the Local Organization committee of EPRV6

2025

Outreach

Outreach talk in Jornadas Espaciais 2026 (Lisbon, Portugal); "Exoplanetas e Mundos distantes"

May 2026

Outreach talk in COMICCON 2026 (Portugal); "PoET - um telescópio solar português"

Apr 2026

Ignite sessions; 'Torres Vedras [↗](#) Ílhavo [↗](#) Armamar'

2023-2024

Outreach talks to highschool students: Espaço vai à Escola - Descoberta de outra Terra - deteção de planetas fora do sistema solar

2022-2025

2022 (3 talks); 2023 (5 online; 5 in-person); 2024 (6 online; 4 in-person); 2025 (6 in-person; 17 online)